

BEE 4750 Lab 2: Monte Carlo

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Due Date

Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

Activating project at `~/hw2-natalie/lab-2-NatalieHo424`

```
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a “seed,” which controls the sequence of numbers which are generated (since these are not *really* random, just “pseudorandom”). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

`TaskLocalRNG()`

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

Seeds and Reproducing Solutions

If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the [Distributions.jl package](#). The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

```
# you don't have to name this "normal_distribution"
#  is the mean and  is the standard deviation
normal_distribution = Normal( , )
# a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for an
uniform_distribution = Uniform(a, b)
```

There are lots of both [univariate](#) and [multivariate](#) distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

```
# draw n samples
rand(normal_distribution, n)
```

Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a [Discrete Uniform distribution](#).

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

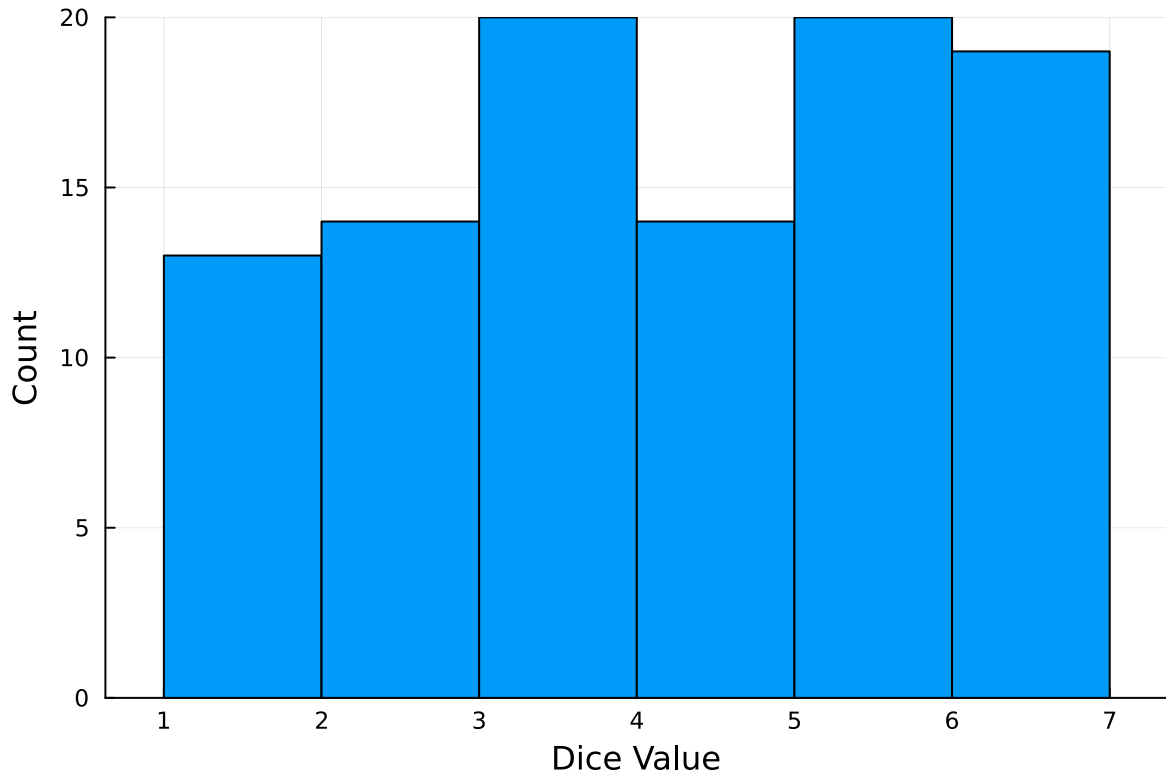
100-element Vector{Int64}:

```
1
3
5
4
6
2
5
5
5
2

3
6
5
5
6
3
6
6
6
```

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim \text{LogNormal}(1.2, 0.3)$;
2. $h \sim \text{GeneralizedExtremeValue}(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

```
using Plots, Random, Distributions

Random.seed!(1)

# Depth-damage parameters
L = 200_000.0
k = 0.8
h0 = 3.0

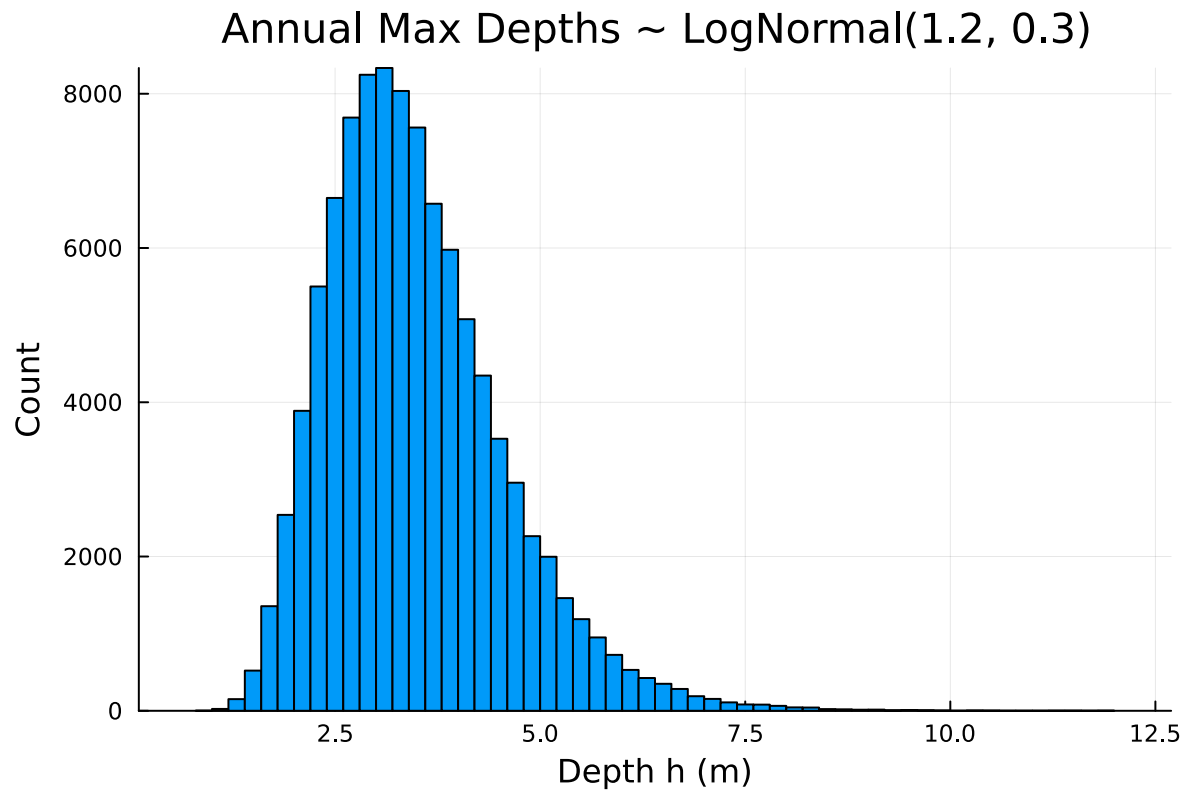
#Logistic Function
d_of_h(h) = (h > 0) ? L / (1 + exp(-k*(h - h0))) : 0.0

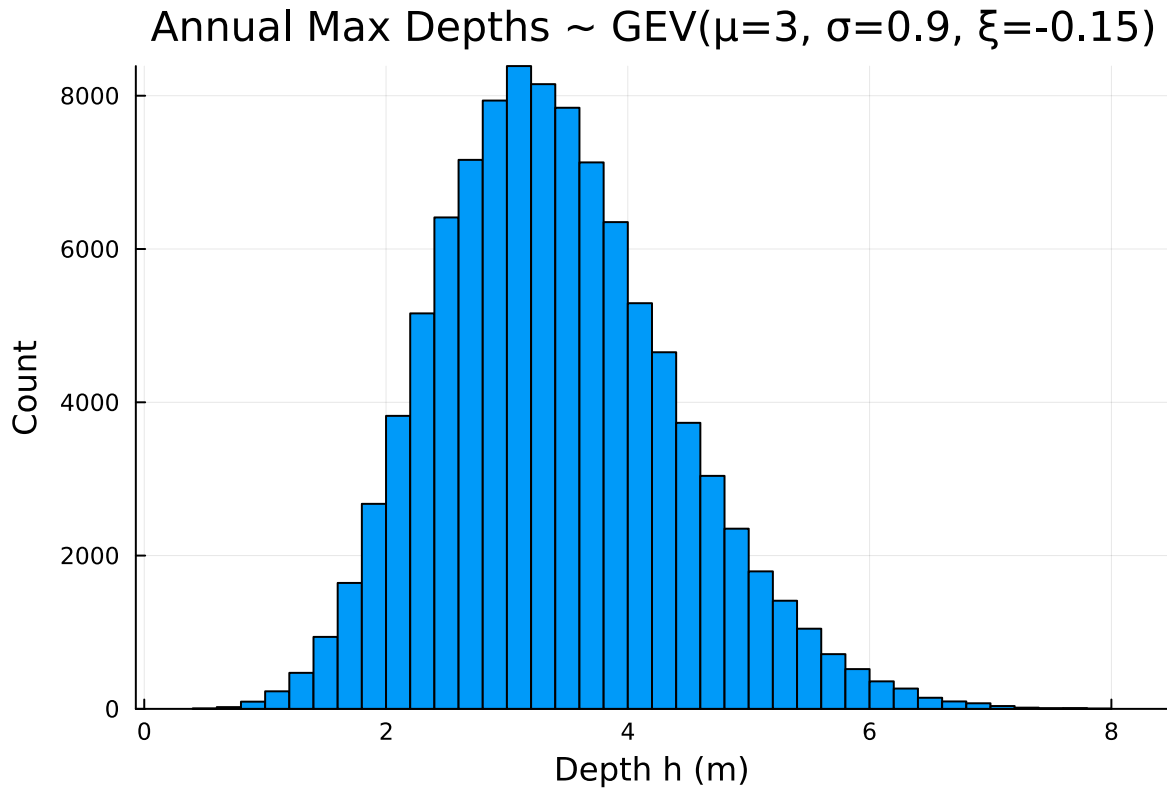
# Hazard distributions
dist_logn = LogNormal(1.2, 0.3)
dist_gev = GeneralizedExtremeValue(3.0, 0.9, -0.15)

# Samples
n = 100_000
h_logn = rand(dist_logn, n)
h_gev = rand(dist_gev, n)
```

```
# Plots
plot1 = histogram(h_logn; bins=60, xlabel="Depth h (m)", ylabel="Count",title="Annual Max Dep
display(plot1)

plot2 = histogram(h_gev; bins=60, xlabel="Depth h (m)", ylabel="Count",title="Annual Max Dep
display(plot2)
```





For plotting the histograms, I began by writing out the parameters provided in the problem and applying the logistic function. I recognized that I also needed to include the value 0 in the function definition; without this, I was receiving an error message. This cutoff is important because the logistic function by itself would otherwise predict small positive damages even when $h \leq 0$, which is physically unrealistic. To generate the plots, I adapted the dice roll example code and applied it to this problem, which allowed me to successfully produce the histograms shown above.

From these results, I observed that the LogNormal distribution appears right-skewed. After doing some research, I learned that this is typical: LogNormal distributions are usually positively skewed with long right tails, which occur due to low mean values and high variances in the random variables.

In contrast, the Generalized Extreme Value (GEV) histogram appears more uniform compared to the LogNormal. This behavior can be anticipated from the given shape parameter, $\xi = -0.15$, which indicates a thinner right tail.

```
using Statistics
```

```
# Convert sampled depths into damages because we are trying to estimate flood damage to a house
d_logn = d_of_h.(h_logn)
```

```

d_gev = d_of_h(h_gev)

# Monte Carlo Summaries
mean_logn = mean(d_logn)
mean_gev = mean(d_gev)

se_mean_logn = std(d_logn) / sqrt(length(d_logn))
se_mean_gev = std(d_gev) / sqrt(length(d_gev))

q99_logn = quantile(d_logn, 0.99)
q99_gev = quantile(d_gev, 0.99)

# Results
println("Monte Carlo results (n = $n)\n")

println("LogNormal results:")
println(" Expected damage (mean) = ", round(mean_logn, digits=2))
println(" Standard error of mean = ", round(se_mean_logn, digits=2))
println(" 99th percentile of damages = ", round(q99_logn, digits=2))
println()

println("GEV results:")
println(" Expected damage (mean) = ", round(mean_gev, digits=2))
println(" Standard error of mean = ", round(se_mean_gev, digits=2))
println(" 99th percentile of damages = ", round(q99_gev, digits=2))

```

Monte Carlo results (n = 100000)

LogNormal results:

Expected damage (mean) = 115307.29
Standard error of mean = 108.48
99th percentile of damages = 190131.39

GEV results:

Expected damage (mean) = 113568.49
Standard error of mean = 106.35
99th percentile of damages = 183428.44

Monte Carlo Analysis of Annual Maximum Flood Damages

I modeled annual maximum flood depths using two candidate distributions:

1. $h \sim \text{LogNormal}(\mu = 1.2, \sigma = 0.3)$
2. $h \sim \text{GEV}(\mu = 3.0, \sigma = 0.9, \xi = -0.15)$

I then mapped each depth sample into a monetary loss using the bounded logistic depth-damage function:

$$d(h) = \begin{cases} \frac{L}{1 + e^{-k(h-h_0)}}, & h > 0 \\ 0, & h \leq 0 \end{cases}$$

with parameters $L = 200,000$ USD (maximum possible loss), $k = 0.8$ (slope), and $h_0 = 3.0$ m (inflection depth).

This function makes sure that damages are zero below the floor and saturate at the maximum loss for very deep floods. This reasoning was provided to us in class.

Results: Monte Carlo Estimates

Using $n = 100,000$ samples from each distribution:

- **LogNormal Characterization**

- Expected annual maximum damage: $\mathbb{E}[D] \approx \$115,307.29$
- Standard error of the mean: $SE(\hat{E}[D]) \approx \$108.48$
- 99th percentile damage: $Q_{0.99}(D) \approx \$190,131.39$

- **GEV Characterization**

- Expected annual maximum damage: $\mathbb{E}[D] \approx \$113,568.49$
- Standard error of the mean: $SE(\hat{E}[D]) \approx \$106.35$
- 99th percentile damage: $Q_{0.99}(D) \approx \$183,428.44$

I see from the results here that both models yield similar estimates. However, the LogNormal Characterization consistently has greater values for the expected annual mean damage, standard error of the mean, and 99th percentile damage. This makes sense because the LogNormal produces slightly higher values because of its heavier right tail.

Choice of Sample Size

I used $n = 100,000$ samples for each distribution to ensure statistical accuracy and computational efficiency. My reasoning is provided below.

The Monte Carlo standard error (SE) of the sample mean is given by

$$SE(\hat{E}[D]) = \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation of damages. I ran the code for calculating this value below. For our simulations, $s \approx 33,631$ but for this case I assumed, $s \approx 34,000$. With $n = 100,000$, the standard error of the mean becomes

$$SE(\hat{E}[D]) \approx \frac{34,000}{\sqrt{100,000}} \approx 108 \text{ USD},$$

which is very small compared to mean damages of up to \$100,000.

I also tested smaller n values to see how this value changes: - At $n = 10,000$, the SE was about \$340, still small but large enough that repeated runs caused noticeable variation in the estimated 99th percentile.

- At $n = 1,000$, the SE grew to about \$1,080, which is way too large for precise reporting.

I then compared these values to the 99th percentile. For the **99th percentile**, I believe that accuracy depends not only on n but also on how many samples actually fall in the upper 1% of the distribution. With $n = 100,000$, I have about 1,000 data points in the top 1%, which provides a stable estimate of $Q_{0.99}$. With $n = 10,000$, only about 100 points fall in the top 1%, which makes the quantile estimate less stable and more sensitive to random variation. Thus, I went with the larger sample size number of 100,000 as I found this to be a better representation of data accuracy for this problem.

In summary, I choose $n = 100,000$ because: 1. It reduces the standard error of the mean to less than 0.1% of the expected damage.

2. It provides enough data in the extreme tail (1,000 values above the 99th percentile) to yield a stable estimate of $Q_{0.99}$.

Thus, I believe this makes $n = 100,000$ a reasonable choice for accurate and efficient Monte Carlo estimation.

Why the Estimates may Differ

From a discussion in class, I decided to answer this question by analyzing the distributions of the plots. The two distributions differ in tail behavior:

- The **GEV distribution** with $\xi = -0.15$ has an upper bound on flood depth at

$$h_{\max} = \mu - \frac{\sigma}{\xi} = 3.0 - \frac{0.9}{-0.15} \approx 9 \text{ m.}$$

This caps the maximum simulated depth.

- The **LogNormal distribution** has an unbounded right tail, allowing for occasional extreme depths.

However, the logistic damage function seems to have an **upper limit** meaning once depths exceed about 5–6 m, the predicted damages approach the maximum loss $L = 200,000$. At this point, additional increases in flood depth produce almost no further change in the estimated damage, since the curve levels off near its maximum value.

As a result: - Both distributions produce very similar expected damages.

- The LogNormal produces slightly higher 99th percentiles due to its heavier tail, but the differences are small because of the damage cap.

Also, the two hazard distributions differ not only in their tails but also in their **overall shape**, which affects how much probability mass falls near the steepest part of the depth–damage curve around $h_0 = 3 \text{ m}$.

- The **GEV distribution** with $\xi = -0.15$ is more concentrated, with an upper bound around 9 m. This means fewer simulated floods fall far into the extreme range, and more probability mass is clustered around its location parameter $\mu = 3 \text{ m}$.
- The **LogNormal distribution** is positively skewed, with a longer right tail and more spread. This allows a higher chance of producing depths just above the inflection point h_0 , where damages rise most rapidly. Even before considering the far tail, this broader shape increases the chance of damaging depths.

Because the depth–damage function grows quickly between 2–5 m and then levels off near $L = 200,000$, the **shape of the distribution around this critical region** has a strong influence on both the expected damages and the estimated 99th percentile. The LogNormal's skew and spread push more samples into this high-sensitivity zone, producing slightly larger damage estimates compared to the GEV.

Deciding Which Distribution is “Better”

To decide on which values fit better, I would want to reference more additional information.

1. **Shape Distribution:** The GEV with $\xi < 0$ imposes an upper bound on flood depth (about 9 m here), which may be realistic if there are physical limits such as levees or watershed capacity. The LogNormal, by contrast, has a positively skewed, unbounded tail, could be more appropriate in settings where extreme depths are theoretically possible.
2. **Extreme Values:** Since damages are highly sensitive to flood depths just above the threshold h_0 , the model that best predicts the upper quantiles (e.g., 95th–99th percentiles) will provide more reliable risk estimates. Past references of historical extreme events would help decide this.
3. **Decision sensitivity:** The choice of distribution matters most if decisions like design standards change depending on the expected loss or high quantile estimates.

Thus, additional hydrologic and site-specific data would be needed to determine which distribution is better. Both models could be reasonable. The GEV is often preferred because it is designed specifically for analyzing annual maximum events, while the LogNormal can be useful when we expect more skewed data and the possibility of very large, unbounded flood depths.

```
std(d_logn)
std(d_gev)
```

33631.28242594318

References

Put any consulted sources here, including classmates you worked with/who helped you.

I referenced the website Statistics How To, to get a better understanding of the GEV distribution because I wanted to know more about what the values provided represent. I learned that the last value provided tells us the shape.